MMSE Pair Scheduling: Computational Complexity and Certified Power Allocation

Pooya Hejazi and Masoud Sabaei

***Abstract*—Uplink pair scheduling couples pairing, resource assignment and powers when the minimum mean-square error (MMSE) receiver changes with the allocation. We consider a common receiver serving every user exactly once, with two users per orthogonal resource, strictly positive bounded transmit powers and a shared budget. We establish fixed-power scheduling hardness with three receive dimensions and rational unit channels from a fixed finite alphabet. A continuous-power extension retains a constant additive separation. We derive the sharp whole-quadrant concavity condition for the pair log-rate and show that rectangular power pricing requires at most seventeen candidate points. We also construct an offline reference with verified continuous upper bounds; its implemented pricing uses rectangular subdivision, while the finite-candidate oracle is tested separately. After a disclosed channel-window correction, we compare methods on 24 eight-user and eight sixteen-user networks in two ray-tracing scenarios. Pricing tightens 11 of 24 main bounds and all eight scale bounds. It does not improve achieved rates over the corresponding unpriced search or yield a 0.01-nat target success. The registered success-fraction difference is zero, with conservative 95% interval $[-0.3449, 0.3449]$. The bounds quantify nominal allocation quality; scalable near-optimality remains unestablished.**

***Index Terms*—MMSE pair scheduling, computational complexity, certified power allocation.**

I. Introduction

UPLINK pair scheduling lets two users share a resource while a linear receiver separates their signals without successive interference cancellation (SIC). Their rates depend on channel alignment and powers. Changing a power also changes the minimum mean-square error (MMSE) filter, coupling pairing and power allocation.

The two-user no-SIC rate is established [1]. Fan et al. [2] optimize MMSE-aware pairing at prescribed powers; Ruder et al. [3] also examine imperfect scheduling information. We ask how the quality of a joint schedule and continuous powers can be bounded for a specified channel realization. We consider a common receiver, two users per orthogonal resource, serving every user exactly once, strictly positive bounded transmit powers and one shared budget.

A feasible schedule gives an achievable rate but not its distance from the optimum. A finite power grid can miss better continuous allocations. We therefore seek both a feasible allocation and a valid continuous upper bound; their difference certifies an upper bound on the optimality gap. We make three contributions:

1) We establish fixed-power scheduling hardness with three receive dimensions and rational unit channels from a fixed finite alphabet. A continuous-power extension retains a constant additive separation.
2) We derive the sharp whole-quadrant concavity condition for the pair log-rate and characterize rectangular pricing by at most seventeen candidates, with an interior polynomial of degree at most five.
3) We construct independently checkable global bounds and compare grid, curvature, pricing and matching–WMMSE methods under common constraints and budgets. Network pricing uses verified rectangular subdivision; the finite-candidate oracle is tested separately. Pricing tightens some bounds, without increasing achieved rates or establishing superiority on the registered accuracy target.

The evaluation uses one eight-antenna receiver in each of two ray-traced scenes. No main or scale case reaches the 0.01-nat target. An original channel-window error and its separately registered correction are disclosed with both sets of outcomes. The accompanying supplementary material supplies the complete proofs, implementation details and experimental records. Throughout, $\ln$ is the natural logarithm and $(\cdot)^H$ the conjugate transpose.

Section II reviews related work. Section III defines the model and allocation problem; Sections IV and V develop the structural results and certified reference algorithm. Section VI reports the corrected experiments, Section VII discusses their limits, and Section VIII concludes. Supplementary Sections S11–S15 collect the full proofs, implemented methods, protocols, outcomes and reproducibility details.

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II. Related Work

A. Pair scheduling and computational complexity

Table I compares the closest receiver, scheduling and optimization precedents. Ruder et al. [4] also distinguish homogeneous-resource pairing from frequency-selective grouping. These works establish the rate and allocation architecture; our question concerns continuous powers and a bound for an integral full-service schedule.

Prior hardness results address distinct allocation models. Prasad et al. [5] study LTE-A multistream scheduling with contiguity and control constraints; Salaün et al. [6] study subcarrier/power allocation with SIC and individual budgets. Mollanoori and Ghaderi [7] consider full-service scalar SIC scheduling with variable group sizes. Razaviyayn et al. [8] establish three-antenna interference-alignment hardness, and Liu [9] establishes two-antenna continuous-transceiver hardness. Together, these results show that full service and a fixed antenna count are not new sources of hardness by themselves. Our question is whether hardness persists with a common no-SIC receiver, exactly two users per resource, three receive coordinates and a fixed finite rational channel alphabet.

B. Power optimization and bounds

Shen and Yu's quadratic transform [10] gives an exact auxiliary reformulation with stationary-point convergence under its assumptions. Their matching-based scheduling [11] and Wang et al.'s weighted-MMSE precoding with error covariance [12] motivate our common-model comparator. The fixed-filter and joint optimization guarantees remain distinct, as Table I explains.

Ul Hassan et al. [13] derive finite candidates for two-link power sharing with fixed interference gains and an exhausted sum budget. Sharma and Teneketzis [14] obtain decentralized guarantees under concave utilities; their MMSE example uses negative MSE rather than our log-rate. We specialize mixed-monotone optimization [15] to the recomputed pair rate. Barnhart et al.'s branch-and-price framework [16] requires complete or bounded pricing. Zhao and Pottie's divisible-spectrum convexification [17] permits sharing unavailable to one integral schedule.

C. Recent resource-allocation designs

Ou et al. [21] transmit two streams per dual-polarized user in a no-SIC uplink, with estimation errors and a finite-blocklength dispersion penalty. Algorithm 1 fixes combiners before the power loop. This directly relevant no-SIC design highlights the receiver assumption important: we recompute MMSE filters as powers change, and optimize integral pair/resource assignments under an aggregate budget.

Chen et al. [22] address beam squint through joint downlink grouping, subcarrier allocation and hybrid beamforming. Their at-least-once service model and changing penalties support a broader design, but do not yield monotonic sum-rate improvement for our exact-once uplink problem. Elyasi and Vosoughi [23] combine multicell MMSE, pilots and powers, using a high-SINR approximation and frozen coefficients for sum SE. Daei et al. [24] use heuristic frame search and projected power ascent under channel aging; their fixed-point convergence result concerns SE evaluation.

Fu et al. [25] optimize downlink power and regularization after large-scale-fading association using a large-system deterministic equivalent and a reported stationarity guarantee, rather than a finite-instance bound. Liu et al. [26] develop permutation-aware mobile-AP learning; permutation guarantees do not establish global reward optimality. Mobini et al. [27] combine heuristic fronthaul-aware association and fixed-association stationarity with MMSE-SIC and per-AP budgets. These studies address mobility, fronthaul, channel uncertainty or downlink beamforming that our nominal pair model leaves outside its scope. They motivate practical extensions; our narrower question is how tightly one integral schedule and its continuous powers can be bounded. The supplement compares each method in detail and identifies the inspected source version.

Our earlier work addresses two complementary parts of this design. Hejazi and Sabaei [28] jointly optimize user clustering and base-station assignment in a user-centric cell-free architecture, then train an ensemble model on optimized assignments. Its NOMA/SIC receiver and multi-AP association differ from the common no-SIC receiver studied here. Hejazi et al. [29] refine an LS channel estimate with a CNN trained using GAN-augmented data to reduce pilot overhead. That estimator supplies upstream CSI; the present bounds apply to fixed nominal channels and do not certify rates under its estimation errors.

III. System Model and Problem Formulation

A. Uplink transmission and information assumptions

We consider $K = 2S$ single-antenna users and a common M-dimensional receiver over S orthogonal resources. Exactly two users share each resource, and every user is served once. This full-service requirement separates the pairing decision from admission control: a user with an unfavorable channel cannot be dropped to improve the objective.

Let $\mathcal{K} = \{1, \dots, K\}$, $\mathcal{S} = \{1, \dots, S\}$, and $\mathcal{E} = \{\{u, v\} : u < v,\ u, v \in \mathcal{K}\}$. The channel of user i on resource s is $\mathbf{h}_{is} \in \mathbb{C}^M$. After whitening by the known positive-definite noise covariance, the received signal for pair $e = \{u, v\}$ is

TABLE I
COMPARISON WITH THE CLOSEST PRIOR WORK.

Work	Contribution and strength	Relevant scope limitation	Question addressed here
Haustein [1]	Derives recomputed-MMSE pair rates and identifies nonconcavity	Receiver analysis does not settle joint schedule/power quality	Restricted hardness, sharp curvature and continuous bounds
Fan et al. [2]	MMSE-aware pairing and binary frequency allocation with branch-and-bound	Prescribed powers and contiguous resource patterns	Positive continuous powers with a shared budget
Ruder et al. [3]	Pairing with predicted/interpolated scheduling CSI and pilot accounting	Fixed powers; perfect payload-equalization CSI	Nominal continuous-power certification, without a mismatch guarantee
Mostafa et al. [18]	Channel-orthogonality reduction and pair/master offloading design	Growing dimension; SIC, admission and delay constraints	Fixed three-dimensional finite-alphabet threshold hardness
Shi et al. [19]	Rate–MMSE relation and globally solved fixed-filter power steps	Alternating joint problem remains nonconvex	Complete pair pricing with the receiver recomputed
Viterbo–Hottinen [20]	Equal-power MMSE pairing; distinguishes disjoint pairs and expanded cyclic schedules	Resource-invariant channels and different scheduling families	Resource-labelled full-service pairs with continuous powers
Shen–Yu [11]	Matching-based joint scheduling, power and beamformer updates	Continuous stationarity for a fixed schedule is not joint global optimality	Verified bounds for an integral common-receiver allocation
Wang et al. [12]	Weighted-MMSE precoding with error covariance and centralized/distributed processing	Statistical/estimated-CSI objective and iterative precoding	Fixed-channel scalar powers and a common-model comparator
Matthiesen et al. [15]	Globally convergent mixed-monotone branch-reduce-bound	Subdivision may be costly; bounds depend on the objective	Pair-specific curvature and pricing within that framework
Ou et al. [21]	SIC-free dual-polarized uplink with estimated CSI and finite-blocklength power control	Fixed power-independent combiners; per-user budgets; conditional power stationarity	Integral pair/resource decisions and bounds with recomputed MMSE

Limitations describe scope of applicability, not measured inferiority.

$$\mathbf{y}_s = \sqrt{p_u}\mathbf{h}_{us}a_u + \sqrt{p_v}\mathbf{h}_{vs}a_v + \mathbf{n}_s, \qquad \mathbb{E}[\mathbf{n}_s\mathbf{n}_s^H] = \mathbf{I}_M. \tag{1}$$

The symbols $a_i \sim \mathcal{CN}(0,1)$ are independent across users and independent of $\mathbf{n}_s \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_M)$. They are decoded with residual interference treated as noise; finite constellations and decoding errors are not modeled. Powers are in watts. For scalar noise, whitening divides each physical channel by the noise standard deviation; path loss already present is not reapplied.

The scheduler and receiver know the same channel realization. We use nominal to mean that optimization and certification refer to this specified realization, including its stored precision. A guarantee under imperfect channel state information (CSI) would additionally require an uncertainty model.

Users satisfy $0 < p_{\min} \le p_i \le p_{\max}$ and a shared budget $\sum_i p_i \le P$. The controller enforces P as an aggregate radiated-power allowance across the scheduled users and resources, in addition to each user's bounds. It does not represent a shared transmitter battery. The positive lower bound prevents removal of service through zero power. Since every pair is eligible, $Kp_{\min} \le P$ guarantees a feasible power allocation for any schedule. Equal-bandwidth resources are mutually orthogonal. We maximize unweighted sum rate without quality-of-service constraints, SIC or additional local power budgets.

Fig. 1 places the full-service architecture in an illustrative physical setting. The evaluated receiver is one eight-antenna AP; the theory allows general M. Pair identities and resource labels are optimization decisions, not fixed sequential-index pairing.

B. Recomputed linear MMSE reception

For each selected pair, the receiver recomputes the linear MMSE filter at the selected powers and treats the other user's signal as noise. We use the established two-user rate derived by Haustein [1]; our analysis concerns its curvature and its role in joint allocation. Define

$$\begin{aligned} g_{is} &= \|\mathbf{h}_{is}\|^2 \ge 0, \\ r_{uvs} &= \frac{|\mathbf{h}_{us}^H\mathbf{h}_{vs}|^2}{g_{us}g_{vs}} \in [0,1] \quad (g_{us}g_{vs} > 0). \end{aligned} \tag{2}$$

Here r_{uvs} is squared normalized channel correlation: zero denotes orthogonal channels and one denotes collinear nonzero channels. If either gain is zero, set

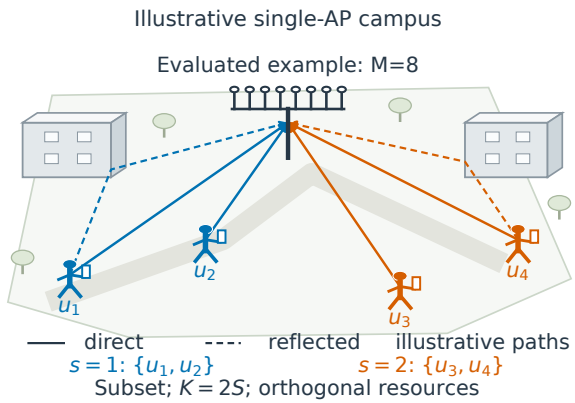


Fig. 1. Illustrative uplink: one eight-antenna AP, pairs on orthogonal resources and linear MMSE without SIC. Each user is served once under positive power bounds and a shared budget. Geometry and propagation paths are schematic, not measurements.

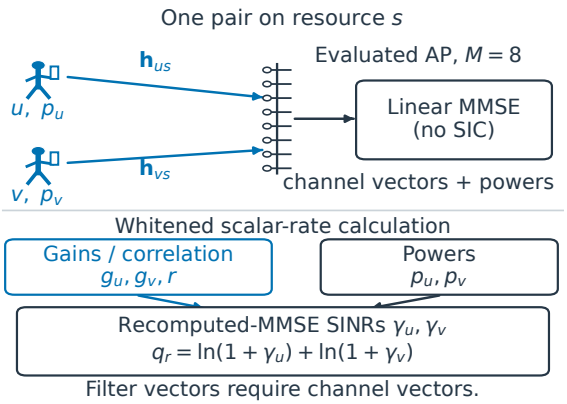


Fig. 2. One pair on a common resource. Whitened gains, correlation and powers determine (3)–(4); constructing receive filters requires full channel vectors. The schematic adds no measured geometry, channel-estimation or SIC model.

$r_{uvs} = 0$; the corresponding zero normalized received power makes this convention consistent with the MMSE rates. The zero-channel user still receives service and minimum power. With $x = g_{us}p_u$ and $y = g_{vs}p_v$, the matrix inversion lemma gives the signal-to-interference-plus-noise ratios (SINRs)

$$\gamma_u = x\frac{1+(1-r_{uvs})y}{1+y}, \qquad \gamma_v = y\frac{1+(1-r_{uvs})x}{1+x}. \tag{3}$$

For generic normalized powers $x, y \geq 0$ and squared correlation $r \in [0,1]$, define the pair sum rate in nats per channel use:

$$\begin{aligned} q_r(x,y) &= 2\ln d - \ln(1+x) - \ln(1+y), \\ d &= 1 + x + y + (1-r)xy. \end{aligned} \tag{4}$$

We write $q_{uvs}(p_u, p_v) = q_{r_{uvs}}(g_{us}p_u, g_{vs}p_v)$. Formula (4) includes receiver recomputation; fixing receiver coefficients while varying powers gives a different objective.

Let $e_s \in \mathcal{E}$ be the pair selected on resource s. The aggregate sum of per-resource spectral efficiencies (SEs) is

$$Q = \frac{1}{\ln 2}\sum_{s=1}^{S} q_{e_s,s}. \tag{5}$$

Thus Q sums bit/channel-use values across resources; gross occupied-band SE is Q/S bit/s/Hz. For resource bandwidth B_r and data fraction η, throughput is $T = \eta B_\mathrm{r} Q$ bit/s and net occupied-band SE is $\eta Q/S$. Neither B_r nor η follows from a channel array alone. System energy efficiency would additionally require amplifier, circuit and processing-power models.

Fig. 2 shows the scalar rate calculation for one pair. Gains and correlation suffice to evaluate the SINRs with MMSE recomputation through (3); constructing the receive-filter vectors themselves requires the channel vectors.

C. Joint continuous problem

Let $z_{uvs} \in \{0,1\}$ indicate the pair placed on resource s. The joint problem is

$$\begin{aligned} \mathcal{P}_\mathrm{C}: \quad \max_{z,p} \quad & \sum_{s\in\mathcal{S}}\sum_{u<v} z_{uvs} q_{uvs}(p_u,p_v) \\ \text{s.t.} \quad & \sum_{s}\Big(\sum_{v<i} z_{vis} + \sum_{v>i} z_{ivs}\Big) = 1 \ (i \in \mathcal{K}), \\ & \sum_{u<v} z_{uvs} = 1 \quad (s \in \mathcal{S}), \\ & p_{\min} \leq p_i \leq p_{\max} \quad (i \in \mathcal{K}), \\ & \sum_i p_i \leq P \end{aligned} \tag{6}$$

The first two equalities enforce service once per user and one pair per resource. Powers can be indexed by users because each appears once, but its rate still depends on its partner and resource. Problem (6) therefore combines discrete scheduling and continuous powers. Even concave pair utilities would leave the scheduling decisions to be resolved.

Viterbo and Hottinen [20] distinguish fixed-power disjoint pairing from an expanded cyclic scheduling family. For our disjoint pairs with resource-invariant channels, the weights q_{uvs} do not depend on s, and maximum-weight perfect matching solves the scheduling step. Edmonds' general-graph matching framework [30] applies; an unrestricted bipartite assignment need not produce

disjoint pairs. Resource-dependent channels require the resource labels to remain in the problem.

D. Finite-grid integer and linear programs

Let $\mathcal{G} \subset [p_{\min}, p_{\max}]$ be a prescribed finite power grid. A configuration $c = (u, v, s, p_u, p_v)$ includes its resource, pair and two powers. Let $\mathcal{C}_{\mathcal{G}}$ contain all such configurations, $a_{ic} = 1$ when user i belongs to c, $b_{sc} = 1$ when c uses resource s, $d_c = p_u + p_v$, and $q_c = q_{uvs}(p_u, p_v)$. The grid integer program (IP) is

$$\begin{aligned} \mathcal{P}_{\mathrm{IP}}(\mathcal{G}): \quad \max_{z} \quad & \sum_{c} q_c z_c \\ \text{s.t.} \quad & \sum_{c} a_{ic} z_c = 1, \quad i \in \mathcal{K}, \\ & \sum_{c} b_{sc} z_c = 1, \quad s \in \mathcal{S}, \\ & \sum_{c} d_c z_c \le P, \qquad z_c \in \{0,1\} \end{aligned} \tag{7}$$

The grid linear program (LP), $\mathcal{P}_{\mathrm{LP}}(\mathcal{G})$, replaces integrality with $z_c \ge 0$. The incidence equalities imply $z_c \le 1$. Its dual uses unrestricted user prices α_i, unrestricted resource prices β_s, and a nonnegative power price λ:

$$\begin{aligned} \min_{\alpha,\beta,\lambda} \quad & \sum_{i} \alpha_i + \sum_{s} \beta_s + \lambda P \\ \text{s.t.} \quad & \alpha_u + \alpha_v + \beta_s + \lambda(p_u + p_v) \ge q_{uvs}(p_u, p_v), \\ & \qquad\qquad c \in \mathcal{C}_{\mathcal{G}}, \\ & \lambda \ge 0 \end{aligned} \tag{8}$$

The grid LP upper-bounds the grid IP, but may lie below the continuous optimum in (6): powers omitted from $\mathcal{G}$ can yield better allocations. A feasible grid IP schedule, in contrast, is feasible for (6) and supplies a lower bound after its rate is validated. Thus the grid gives useful candidates without by itself certifying the continuous optimum.

Replacing grid weights by nonnegative Borel measures on each pair's power rectangle gives the continuous configuration LP: incidence masses equal one, the total power moment is at most P, and the objective integrates q_{uvs}. Its formal primal is (D9) in Supplementary Section S11; its dual imposes (8) at every power. Write V_{CLP} for its value and V_{C} for (6). Dirac masses embed integral schedules, so $V_{\mathrm{C}} \le V_{\mathrm{CLP}}$; assignment and power mixtures can make the inequality strict.

IV. Structural Analysis

A. Hardness with a fixed receiver dimension

Decision problem IDEAL-MMSE-PAIR. Fix the rational unit-vector alphabet $\mathcal{A} \subset \mathbb{Q}^3$ constructed in Supplementary Section S1. An instance gives $K = 2S$ and an explicit $K \times S$ table of alphabet indices, with unit powers and unit white-noise variance. Does a full-service pair/resource schedule attain at least $K \ln 2$? The threshold is specified by K, not a numerical logarithm. Malformed encodings are rejected.

Theorem 1. IDEAL-MMSE-PAIR is NP-complete. Thus fixed-power sum-rate maximization is NP-hard with $M = 3$ and the fixed rational unit alphabet $\mathcal{A}$. For fixed rational $0 < p_{\min} < \bar{p} < p_{\max}$, the continuous-power family with $P = K\bar{p}$ is also NP-hard through ideal-threshold attainment. Both constructions have a positive constant additive separation in natural-log sum-rate units.

Proof sketch. Use perfect 3DM with each element in at most three triples [31, Sec. 2, (2.1)]. Users are $X \cup Y$ and resources are Z; each resource's admissible graph has at most three edges and is a forest. Section S1 realizes exactly these edges as zero inner products, including an outsider role. A unit-power pair reaches $2 \ln 2$ exactly when orthogonal. A schedule witness has $O(K \log K)$ bits; the rational product $\prod_i (1 + \gamma_i) \ge 2^K$ is checked with $O(K)$-bit numerator and denominator because the alphabet is fixed. For continuous powers, Jensen bounds the ideal sum by $K \ln(1 + \bar{p})$; a forbidden pair has a fixed positive loss. Section S11 gives reduction size and explicit gaps (D1)–(D3). □

The contribution is the conjunction of restrictions in Theorem 1. The channel-orthogonality and fixed-antenna precedents discussed in Section II remain essential context.

Theorem 1 gives neither a constant relative approximation gap nor NP-membership for unrestricted continuous thresholds. Dimension three is sufficient, not a tractability boundary. Channels vary independently across resources; prescribed common-ray geometry is not asserted. AIM [32, Th. 3.16, p. 1639] supplies the real tree-complement representation antecedent; Section S1 supplies the rational unit alphabet for the bounded forest family.

B. Sharp pairwise curvature

Theorem 2. The function q_r in (4) is jointly concave over $\mathbb{R}^2_+$ if and only if

$$r \le 1 - \frac{1}{\sqrt{2}}. \tag{9}$$

Within this range $\nabla^2 q_r(x, y) \prec 0$ at every finite $(x, y) \in \mathbb{R}^2_+$.

Proof sketch. Let $a = 1 - r$. At $x = 0$, $q_{xx} \to 1 - 2a^2$ as $y \to \infty$, proving necessity by continuity to an interior point. For $a \ge 1/\sqrt{2}$, both diagonals are nonpositive and the determinant numerator has nonnegative coefficients

with positive constant term. Thus the Hessian is negative definite. Supplementary (D4)–(D5) give the Hessian and numerator; Section S2 lists every coefficient. □

This squared-correlation threshold applies to the whole nonnegative quadrant. A finite rectangle can remain concave beyond it; failure of global concavity does not imply positive curvature at every operating point.

C. Complete rectangular power pricing

To extend a configuration dual from grid points to continuous powers, we must bound the rate minus its power cost over every pair rectangle. We call this maximization power pricing:

$$\begin{aligned} H = \max_{(x,y)\in\mathcal{B}} \{q_r(x,y) - \lambda_1 x - \lambda_2 y\}, \\ \mathcal{B} = [L_x, U_x] \times [L_y, U_y]. \end{aligned} \tag{10}$$

For positive gains and a network price λ in natural-log rate units, $\lambda_1 = \lambda/g_{us}$ and $\lambda_2 = \lambda/g_{vs}$. The following result also allows signed prices, although the network power multiplier is nonnegative.

Theorem 3. For $0 < r \leq 1$ and a compact nondegenerate rectangle in $\mathbb{R}^2_+$, the maximum in (10) occurs among at most seventeen points: four corners, at most two stationary points on each of four edges, and at most five interior stationary points.

Proof sketch. For $B(x), C(x)$ explicitly defined in supplementary (D6), the first stationarity equation is $C + By = 0$. For $r > 0$, $B = C = 0$ is impossible at feasible x. Substitution of $y = -C/B$ gives the degree-at-most-five polynomial $\Pi(x)$ in (D7), with $\Pi(-1) = 2r^3 > 0$. Each edge has the nonzero degree-at-most-two polynomial (D8). Four corners, eight edge candidates and five interior candidates suffice; Section S11 gives admissibility and degeneracies. □

For $r = 0$ the objective is separable. For $j = x, y$, set $(\lambda_x, \lambda_y) = (\lambda_1, \lambda_2)$ and use $\mathrm{clip}(1/\lambda_j - 1, [L_j, U_j])$ if $\lambda_j > 0$, or U_j otherwise, where $\mathrm{clip}(t, [L, U]) = \min(U, \max(L, t))$. Collapsed rectangles reduce to an edge or point. A zero-gain user's physical contribution is $-\lambda p_i$, maximized at minimum power for $\lambda \geq 0$; the other coordinate is a scalar logarithm. Additional coupled pair constraints require their own boundary analysis.

Theorem 3 gives finitely many candidates, not a radical formula or precision-independent runtime. Certification requires all relevant real roots and directed logarithmic enclosures; unresolved enclosures are failures. The objective distinguishes this result from earlier elimination methods [13].

V. Certified Network Optimization

A. Repairing a numerical configuration dual

Numerically computed prices may satisfy only the configurations present in a grid master. The following repair raises each resource price enough to cover every continuous configuration, producing a valid bound even when those numerical prices are not optimal.

Proposition 1. Let α_i be arbitrary real values and $\lambda \geq 0$. Suppose H_{uvs} is a valid upper bound on

$$\max_{p_u, p_v \in [p_{\min}, p_{\max}]} \{q_{uvs}(p_u, p_v) - \lambda(p_u + p_v)\}. \tag{11}$$

Set $\beta_s = \max_{u<v}(H_{uvs} - \alpha_u - \alpha_v)$. Then

$$\begin{aligned} U_{\mathrm{dual}} = \sum_i \alpha_i + \sum_s \beta_s + \lambda P, \\ V_{\mathrm{C}} \leq V_{\mathrm{CLP}} \leq U_{\mathrm{dual}}. \end{aligned} \tag{12}$$

Proof. The choice of β_s makes every continuous configuration inequality in (8) valid. Multiply those inequalities by any feasible configuration mass and sum. User and resource equalities produce the first two sums in (12), while nonnegativity of λ bounds the power contribution by λP. This is weak duality and does not require that the candidate prices solve the numerical master exactly. □

Every pair/resource rectangle must be covered, including pairs absent from the numerical master. Let L be a validated lower enclosure of a feasible integral schedule's objective. Then $L \leq V_{\mathrm{C}} \leq U_{\mathrm{dual}}$ bounds its loss by $U_{\mathrm{dual}} - L$. Throughout, U is the smallest retained validated global upper bound, including the fallback U_0, a complete spatial bound and U_{dual} when available. A valid $[L, U]$ need not meet a requested gap tolerance.

Dual repair does not remove the configuration integrality gap. Section S3 gives assignment and fixed-pair power examples. In the latter, $g_u = g_v = r = 1$, $p_i \in [1, 2]$ and $P = 11/5$. The deterministic optimum is $\ln(128/55)$, but masses $4/5$ at $(1, 1)$ and $1/5$ at $(1, 2)$ have the same average power and strictly larger objective. Thus assignment branching alone need not give integral-continuous convergence.

B. Complete small-network reference

There are $K!/2^{K/2}$ assignments of unordered user pairs to labelled resources. Enumerating this family gives a complete reference for small networks, but its growth limits scale. For each assignment, the remaining domain is a box in K physical power variables intersected with $\sum_i p_i \leq P$. Subdividing these powers, rather than only selecting assignments, allows both assignment and power-mixing gaps to be resolved.

For a normalized pair box $[\ell_x, u_x] \times [\ell_y, u_y]$, let $a = 1 - r$. The mixed-monotone bound is

$$U_{\mathrm{MM}} = \ln\left(1 + u_x \frac{1 + a\ell_y}{1 + \ell_y}\right) + \ln\left(1 + u_y \frac{1 + a\ell_x}{1 + \ell_x}\right). \tag{13}$$

Each term increases in its own power and decreases in the interfering normalized power, proving the bound directly from (3). A budget-aware alternative decomposes $q = f + g_x + g_y$, where $f = 2\ln d$, $g_x = -\ln(1 + x)$ and $g_y = -\ln(1 + y)$. The Hessian of f is

$$\nabla^2 f = -\frac{2}{d^2}\begin{bmatrix}(1 + ay)^2 & r \\ r & (1 + ax)^2\end{bmatrix}. \tag{14}$$

The matrix inside (14) is positive semidefinite because $(1 + ax)(1 + ay) \geq 1 \geq r$. Hence f is concave and its tangent majorizes it. Both g_x, g_y are convex and lie below their endpoint chords. The tangent plus both chords is an affine upper bound; degenerate intervals use their point value.

Summing these affine bounds over the assignment and maximizing over the physical power box and total budget is a continuous knapsack problem: start at the lower powers, allocate remaining budget to positive affine slopes in descending order, and stop at each upper bound. The resulting upper bound retains the total-power constraint, unlike independently maximizing every pair. Taking the minimum of it and the summed bounds (13) preserves validity. The tangents, secants and knapsack arithmetic must be evaluated with outward bounds.

When the exact rational condition $(1 - r)^2 \geq 1/2$ holds, Theorem 2 supplies a supporting tangent for the entire pair rate. The accelerated reference uses this tighter affine bound in place of the tangent-secant construction for that pair, retaining the same budget maximization and mixed-monotone intersection. For all remaining pairs it retains the generic bound.

The reference first covers every assignment with a valid upper bound. It then selects a live box with largest bound and bisects a longest physical-power interval. Boxes whose lower powers violate the budget are infeasible; boxes whose upper bounds cannot improve the incumbent are discarded. The lower corner of every retained box is feasible, and additional feasible points may improve the incumbent. At interruption, the maximum live upper bound, combined with the incumbent, remains a valid global interval. The separately computed U_{dual} may be intersected with this upper bound without changing feasibility or completeness.

Proposition 2. In exact arithmetic, with finitely many assignments and the stated compact power domain, best-bound selection with longest-edge bisection and the stated bounds terminates at any tolerance $\varepsilon > 0$ if computation is unrestricted and each retained box's feasible lower corner is evaluated.

Proof. On the compact domain, the bound minus the feasible lower-corner value vanishes uniformly with box diameter. Longest-edge bisection has only finitely many nodes above any positive diameter threshold. Thus only finitely many selected boxes can have bound exceeding $L + \varepsilon$; best-bound selection reaches $\max U_{\mathrm{live}} \leq L + \varepsilon$. Supplementary Section S11 details feasibility and this finite-tolerance argument. □

Proposition 2 establishes convergence, not scalable complexity. Finite precision imposes an arithmetic floor, which must be reported if it prevents the requested enclosure. The pair oracle does not remove scheduling hardness or establish a speedup. The evaluated implementation prices complete rectangular partitions; the algebraic oracle is examined separately.

For initialization, pair users $(2s - 1, 2s)$ on resource s and set every power to $p_0 = \min(p_{\max}, P/K)$. This is feasible when $Kp_{\min} \leq P$. The common fallback is

$$U_0 = \sum_{i=1}^{K} \max_{s \in \mathcal{S}} \ln(1 + g_{is} p_{\max}), \qquad g_{is} = \|\mathbf{h}_{is}\|^2.$$

It drops interference, the aggregate budget and resource-incidence constraints. This loose ceiling remains valid when a tighter proof is unavailable.

A separate checker reconstructs gains and correlations from stored binary64 channel entries, validates schedule/power constraints, and encloses logarithms with integer arithmetic and explicit remainders. Exported partition trees and dual weights establish complete coverage; unvisited assignments retain the interference-free fallback. Recorded verification times determine deadline attainment independently of later worker returns. The supplement describes replay and reuse of identical inputs. Exactness concerns the stored nominal channels.

TABLE II
CERTIFIED OPTIMIZATION PROCEDURE

1. Initialize sequential pairs with feasible equal powers and an interference-free ceiling.
2. Solve the numerical grid IP/LP; validate any improved integral incumbent with outward rate bounds.
3. In the full arm, allocate at most one quarter of the supplied continuation time to price all pair/resource rectangles and repair the dual. Incomplete pricing retains the valid fallback.
4. If time remains, cover every labelled assignment with spatial bounds, adding pair tangents where Theorem 2 applies. Retain the fallback until coverage is complete.
5. Refine the live box with largest bound; stop refining infeasible or dominated boxes while retaining proof coverage. At interruption retain the timely incumbent and a global upper bound, intersected with any complete repaired dual.

C. Evaluation quantities and comparison contract

We report validated feasible values $L/\ln 2$, global gaps $U - L$ in nat, radiated power, deadline attainment and failures. Improving a feasible value increases achieved rate; reducing an upper bound improves knowledge of allocation quality. These are separate outcomes. Comparators use the same channels, receiver, service constraints, power domain and scheduling family, with schedule search included in their computational allowance.

VI. Experimental Evaluation

A. Physical inputs and comparison protocol

We use DeepMIMOv3 0.2.9 geometry-based channel generation [33] to evaluate complete schedules in two ray-tracing scenarios: O1_60 at 60 GHz and ASU_Campus1 at 3.5 GHz. In both scenes we use AP 1 with the DeepMIMO array shape $[1, 8, 1]$ in X, Y, Z order: an eight-element Y-axis array with half-wavelength spacing, zero configured rotation and isotropic elements. Users have one isotropic element. Channels retain up to ten propagation paths and their path loss. Sections S13 and S15 give the settings, frozen source/input hashes and per-case provenance. We assume ideal synchronization, block-static channels and no inter-resource interference or estimation error. The two bands provide contrasting propagation examples; they do not constitute standardized sixth-generation deployment models.

The results below come from a separately registered correction to the original study. Its 32-point, 50-MHz delay window had clipped every retained ASU path, producing zero channels. The original outcomes, including the null primary comparison, remain in the supplement. Before generating new outcomes, we specified a 512-point transform, 32 tones at indices $0, 16, \ldots, 496$, and a 512-sample cyclic prefix (CP). The useful symbol and CP each last $10.24\,\mu$s, and all retained absolute delays fit within the guard. The factor $\sqrt{512}$ reverses the generator's $\sqrt{P_\ell/512}$ path-amplitude normalization (Section S15). The tone spacing is 97.65625 kHz; the four or eight optimized tones occupy 0.390625 or 0.78125 MHz in aggregate, respectively. Noise at -174 dBm/Hz with a 7-dB noise figure gives 1.9485×10^{-15} W per tone. Users are neither delay-shifted individually nor filtered by channel strength.

We report gross occupied-tone spectral efficiency (SE), Q/S. For a validated lower bound L in natural-log sum-rate units, $\mathrm{SE} = L/(S \ln 2)$. The assumed CP would halve payload SE before any other overhead. Accordingly, these values are neither net SE nor throughput over the full 50-MHz band. The guard is an idealized waveform assumption; we do not perform a time-domain validation.

TABLE III
Common-model methods. Inputs and constraints agree within each size; the charged allowance is 18 s at $K = 8$ and 28 s at $K = 16$.

Arm	Feasible search	Continuous upper bound
Grid	Grid ILP and LP	Interference-free ceiling
Curvature	Grid; complete scheduling family; bounded power refinement	Spatial plus eligible pair tangents
Full	Same search; additional box pricing	Spatial/dual intersection
Matching–WMMSE	Alternating matching and continuous powers; exchanges	Common ceiling
Grid+pricing	Grid incumbent, at $K = 16$	Repaired dual after all prices

The corrective pools use previously untouched O1 rows 321–340 and ASU rows 21–40. Each scene supplies 12 eight-user main cases ($S = 4$) and four sixteen-user scale cases ($S = 8$). Fixed seeds select distinct users within a case, with replacement across cases from the same scene pool. All 32 cases are nontrivial, although individually weak or zero channels remain. Powers lie in $[0.05, 0.2]$ W with total budget $K/10$ W. The computational budget determined sample sizes; no prospective statistical power calculation was used. Inference is conditional on these two scenes.

Table III separates each method's feasible search from its upper bound. Grid uses $\{0.05, 0.1, 0.2\}$ W; its integer linear program (ILP) and LP relaxation alone do not bound the continuous optimum. Matching–WMMSE combines resource-labelled matching, receiver/weight updates and shared-budget amplitude updates, with nine starts and bounded exchange refinement. Supplementary Algorithm S1 and (M1)–(M4) specify the implemented updates and stopping rules. Its scheduling step draws on Shen and Yu [11], and its weighted-MMSE updates on Wang et al. [12]. Both are adapted to the present full-service model, so the comparator does not reproduce either source's full system or reported curves. Eight-user runs receive 18 s with a 20-s external watchdog; scale runs receive 28 s with a 30-s watchdog. The allowance includes startup, search, separate certificate checking and persistence of the verified result. The implementation reserves the final 5 s for checking and persistence. Separate pricing and spatial continuations each allow at most 1,200 splits; the spatial root cap is 3,000, covering the 2,520 labelled assignments at $K = 8$ when enumeration completes. These work caps can cause early return; equal allowances do not imply equal consumed work. Section S12 gives the solver and stopping settings. Full uses validated rectangular subdivision for pricing; the seventeen-candidate characterization is examined separately and

does not explain the measured network runtimes. Runs are serial on a recorded nondedicated Intel Core i7-14700HX Windows host, with 20 physical/28 logical cores, about 64 GiB RAM and single-threaded numerical linear algebra. Section S15 records software versions and unrecorded historical fields.

The registered primary endpoint compares full with curvature: the equal-scene-weighted difference in the fraction of runs with a completed verified receipt giving gap at most 0.01 nat by the charged deadline and normal worker return by the external deadline. The target corresponds to gross SE gaps of 0.00361 bit/s/Hz at $S = 4$ and 0.00180 at $S = 8$. It is a numerical accuracy target, not a quality-of-service requirement; it becomes stricter per occupied tone as S increases. Four 98.75% Clopper–Pearson intervals [34] for scene-specific discordant probabilities give a conservative at-least-95% Bonferroni interval [35] for that difference. Primary status accounting retains every planned case; a missing paired status makes its scene estimate undefined and its scene interval $[-1, 1]$. The overall interval averages the two scene intervals. Secondary intervals use 10,000 scene-stratified paired percentile bootstrap resamples [36] of complete valid scalar pairs, with equal scene weights. All 24 main pairs are valid here (12 per scene); missing secondary values would not be imputed as zero. These intervals are exploratory. Original and corrective outcomes are analyzed separately; the supplement gives both protocols, the eight analytic reliability executions, mechanism controls and postprocessing deviations.

Writing $\hat{p}_{10,s}$ and $\hat{p}_{01,s}$ for the two scene-specific discordant fractions, the estimand estimate is $\hat{\Delta} = \frac{1}{2}\sum_{s\in\{\mathrm{O1},\mathrm{ASU}\}}(\hat{p}_{10,s} - \hat{p}_{01,s})$. Section S15 gives the exact interval construction and resampling order.

B. Global bounds and execution reliability

All 128 corrective executions (96 main, 24 scale and eight analytic reliability runs) returned normally, without recorded solver or storage errors or verification timeouts. A separate offline checker reconstructed the accepted bounds from the saved evidence. Automatic postprocessing exceeded its initial allowance, so one disclosed 92.89-s saved-data reproduction completed the analysis and replay without rerunning a performance cell. This extra cost is separate from the charged method durations. The supplement reports 700 published verification receipts, including 417 offline cache hits for identical case/schedule/proof inputs after individual integrity and timing checks. A receipt is an immutable parent record of a completed check; it is not necessarily a distinct mathematical reconstruction.

Fig. 3 shows all 12 main cases per scene. Every method returns normally, but none reaches the 0.01-nat target. Full and curvature therefore have zero successes and 12 joint failures in each scene, giving a primary difference of zero with conservative 95% interval $[-0.3449, 0.3449]$. This establishes neither superiority nor equivalence. Each execution supplies valid bounds, but their gaps exceed the target.

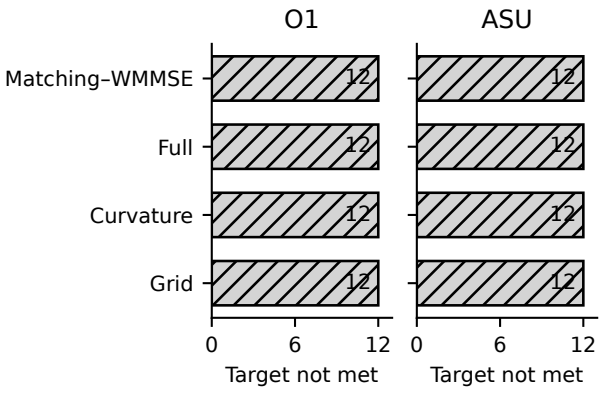


Fig. 3. Operational target attainment: 12 corrective eight-user cases per scene and arm, 18-s allowance. All return normally with valid bounds but miss the target; the zero counts of target successes, errors and missing outcomes are stated explicitly.

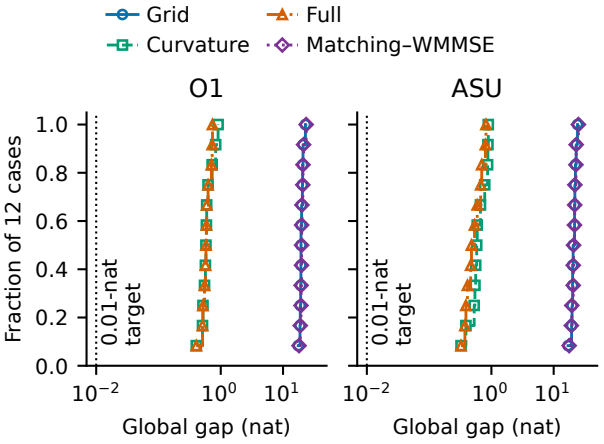


Fig. 4. Continuous global-gap distributions with all 12 cases per curve. The labelled dotted line marks the 0.01-nat certification target. Leftward shifts indicate tighter bounds, not higher achieved rates.

Fig. 4 shows the empirical cumulative distribution of verified global gaps: higher curves at a fixed horizontal value mean more cases with no larger gap. The logarithmic axis separates the grid ceiling from tighter continuous bounds. Mean gaps for grid, curvature and full are respectively 19.784, 0.617 and 0.592 nat in O1, and 20.970, 0.634 and 0.547 nat in ASU. The equal-scene full-minus-curvature mean is -0.0554 nat, with exploratory paired-bootstrap interval $[-0.0851, -0.0295]$. Pricing reduces some upper bounds without creating a target certificate. These gaps bound unknown regret; they are not measured optimality errors.

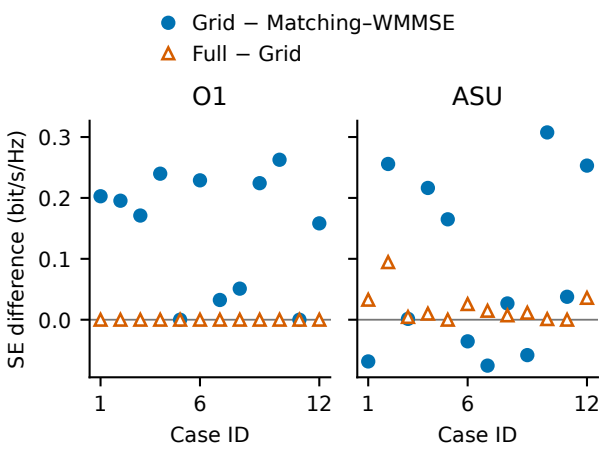


Fig. 5. Paired gross occupied-tone SE differences in the 24 corrective main cases. Zero denotes equal achieved rates. Case IDs are fixed identifiers, not a time or parameter sweep. Full and curvature coincide.

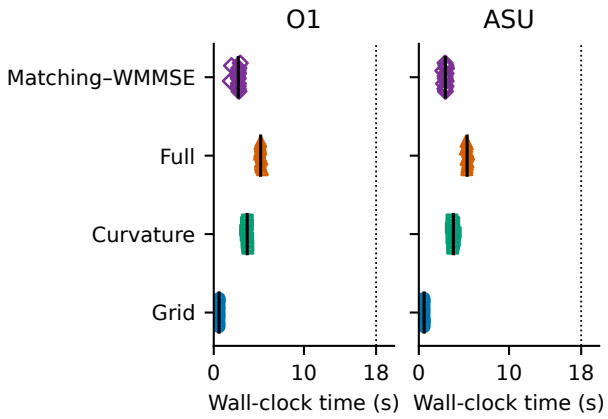


Fig. 6. All main-case parent-process wall-clock durations, including startup, search, checking and persistence; black bars show medians, not confidence intervals. The dotted line is the 18-s charged cap. Every observed return is normal. Measurements concern one nondedicated host.

Fig. 5 separates achieved-rate gains from bound tightening. Grid exceeds matching–WMMSE by a mean 0.1472 bit/s/Hz in O1 and 0.0855 in ASU. Full and curvature achieve identical rates: both equal grid in O1, while their mean ASU improvement over grid is 0.0198 bit/s/Hz. The extra ASU rate therefore comes from their shared refinement. The equal-scene full-minus-matching–WMMSE mean is 0.1263 bit/s/Hz, with exploratory interval $[0.0766, 0.1768]$. This comparison concerns the stated implementations, inputs and budgets; it does not rank the broader methods across other models.

Fig. 6 reports parent-process wall-clock duration, including startup and checking. The full arm averages 5.20 and 5.34 s in O1 and ASU, versus 3.68 and 3.81 s for curvature. Grid averages about 0.58 s and matching–WMMSE 2.60 and 2.93 s. Pricing therefore adds cost in this implementation. The methods stop at their declared finite work caps, well before the 18-s ceiling; these are not time-to-convergence or time-to-target measurements. Early normal return should not be interpreted as a real-time guarantee.

C. Controlled mechanism and fairness

Fig. 7 uses the eight prespecified synthetic four-user cases from the original study, reused once rather than regenerated. Reference, curvature and full continuations begin from the same incumbent and dual, with equal 2-s search caps followed by separate verification. All meet the target; eligible tangents tighten several low-alignment, high-gain cases. Full's repaired dual never improves the final spatial bound in these eight cases. The figure therefore isolates a curvature mechanism but does not demonstrate an incremental pricing benefit.

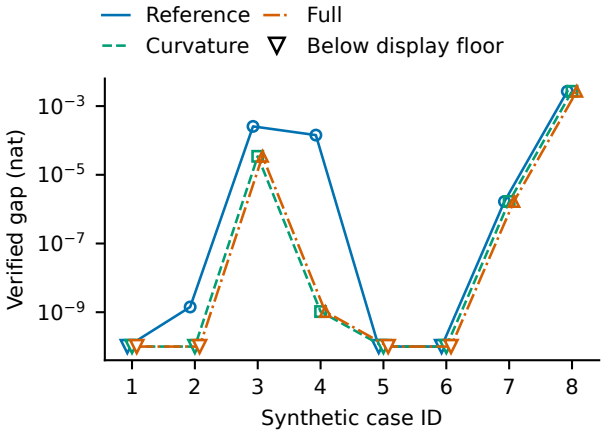


Fig. 7. Same-incumbent mechanism controls, eight original-study synthetic cases and 2-s searches plus verification. The logarithmic display floor is 10^{-10} nat; downward open triangles mark values below that floor, not exact zeros. These are separate from the corrective ray-traced primary population.

Values below 10^{-10} nat share a display floor; they are enclosures, not exact zero errors. The supplement defines the Gaussian mixture, gain scales and seeds. In the main ray-traced cases, only 1.12% of O1 and 1.79% of ASU candidate pair/resource combinations satisfy the global concavity threshold, averaged across cases. These fractions describe where the tangent bound applies; they do not establish its causal contribution to runtime.

For user i, let $R_i = \log_2(1+\gamma_i)/S$ denote its contribution to gross SE over the total occupied bandwidth. We report Jain's index [37], $J = (\sum_i R_i)^2/(K\sum_i R_i^2)$, and summarize $\min_i R_i$ for the comparisons below; the latter is undefined if every rate is zero. Fig. 8 compares user-rate fairness. Higher Jain indices indicate more balanced

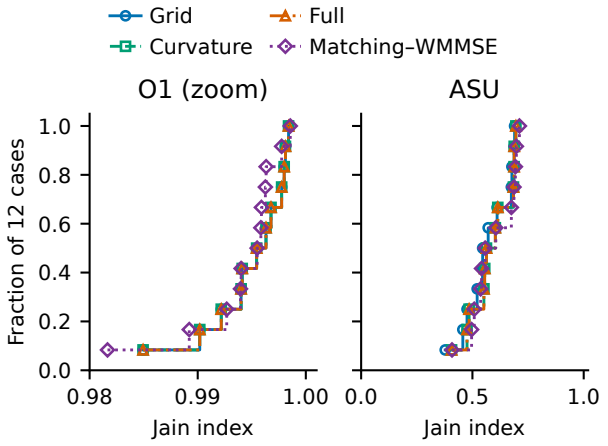


Fig. 8. Jain-index distributions over 12 cases per scene, normalized by that registered count. The O1 panel magnifies its near-one range; complete minimum-rate/fairness summaries are in Section S15. Undefined all-zero values would remain missing. Fairness is descriptive, not the optimization objective.

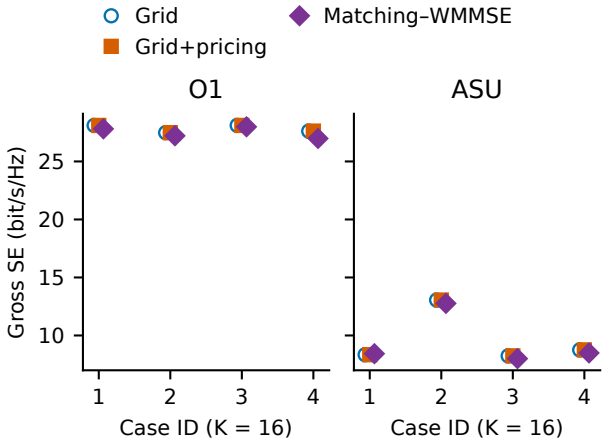


Fig. 9. Gross occupied-tone SE for all eight sixteen-user cases, 28-s allowance per arm. Grid and Grid+pricing have equal values; a small horizontal display offset and open/filled markers expose their overlap. Markers compare methods at each fixed case ID; the IDs do not form a parameter sweep.

rates, but sum-rate maximization imposes no fairness objective. Full has mean index 0.9947 in O1 and 0.5840 in ASU, while matching–WMMSE gives 0.9940 and 0.5930. ASU thus exposes a rate/fairness tradeoff rather than a uniform benefit. Full's mean minimum-user SE is 0.1530 bit/s/Hz in ASU, versus 0.1958 for matching–WMMSE, although both serve every user with positive power. Positive power and full service do not ensure comparable rates for weak users.

D. Sixteen-user results and limits

Fig. 9 retains all four scale cases in each scene. Grid and grid-plus-pricing have identical rates, averaging 27.827 bit/s/Hz in O1 and 9.598 in ASU; matching–WMMSE averages 27.495 and 9.423. The grid accounts for the observed rate advantage. The scale pricing arm avoids enumerating the complete scheduling family, so it is distinct from the eight-user spatial method. Four cases per scene do not establish a scaling law.

Fig. 10 shows relative global gaps, $100(U-L)/L$ for $L>0$; smaller bars mean less uncertainty about the optimum. Since $L \le V_{\mathrm{C}} \le U$ and $L>0$, $(V_{\mathrm{C}}-L)/V_{\mathrm{C}} \le (U-L)/L$; this is a conservative relative-regret bound. Pricing covers all 960 pair/resource rectangles in each scale case. Mean absolute gaps decrease from 38.863 to 7.225 nat in O1 and from 38.580 to 3.933 nat in ASU. Mean relative gaps decrease from 25.20% to 4.68% and from 74.58% to 7.47%, respectively. None reaches 0.01 nat. Thus all eight observed scale bounds improve, while achieved rates remain unchanged and a substantial global gap remains.

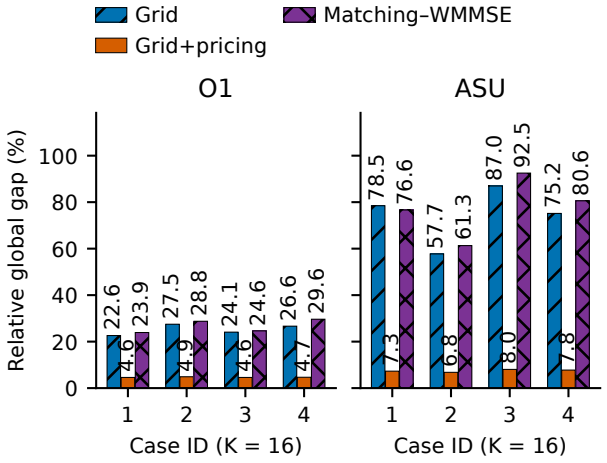


Fig. 10. Relative global gaps in all sixteen-user cases. These percentages bound unknown regret and are not measured rate losses. Bar labels give the displayed percentages. All pricing runs cover 960/960 rectangles; none meets the absolute target.

VII. Limitations and Discussion

Pricing tightens bounds without increasing achieved rate or meeting the target. The complete small-network reference therefore supports offline assessment, not scalable near-optimal or real-time operation.

Separate CSI and GPU studies (Supplementary Sections S6 and S5) limit broader claims. The 120-network, four-user CSI study establishes neither CNN superiority nor equivalence at four pilots; its normalized units differ from the physical-watt study. GPU coefficient construction is faster only at the largest tested size, while GPU LP optimization shows no speedup and produces infeasible iterates at 16 and 32 users. These findings support no

learned-CSI, robustness or end-to-end GPU acceleration guarantee.

VIII. Conclusion

Full-service MMSE scheduling remains hard with three receive dimensions and a finite rational channel alphabet. Sharp concavity and finite rectangular pricing support continuous bounds, while assignment and power mixing explain why the configuration relaxation need not be exact.

Pricing tightens 11 of 24 main upper bounds and all eight scale bounds, without additional rate over the corresponding search without pricing. Separate replay reconstructs the accepted bounds. No corrective main or scale case meets 0.01 nat, and the registered comparison establishes neither superiority nor equivalence. Further work should improve bound refinement, test broader propagation populations and develop channel-uncertainty models.

Acknowledgment

OpenAI Codex (GPT-6-family models, including Astra) assisted mathematical derivation, proof drafting and algebraic checks in Sections III–V and Supplementary Sections S1–S3 and S11; experimental, analysis and certificate-checking code for Section VI and Supplementary Sections S4–S8 and S12–S15; and literature synthesis in Section II and Supplementary Section S9. It also assisted drafting and editing throughout the article and supplement, and code for the schematic diagrams, tables and plots. The accompanying evidence states the checks performed and their limits. The authors remain responsible for the content.

Data Availability

Local artifact PDP2-CR030-R1 accompanies the paper with code, stored cases, verification receipts and replay instructions (Supplementary Sections S10 and S15). Its scientific payload is fixed by `REPRODUCTION_MANIFEST.json` and SHA-256 hashes; the delivery manifest binds the article and archives. Raw DeepMIMO scenarios are excluded. Public access to the revised repository release remains pending verification.

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